SOFTWARE UART FOR SHARP PC-1500(A) by Alex F4VTS

SERIN and SEROUT routines

This software creates a 'Software UART' in the PC-1500(A) pocket computer without a dedicated interface (such as the CE-158) and without any hardware modification*.

**However, a simple, optional modification is described that lets you use the PC-1500(A) docked in the CE-150 printer/cassette interface at the same time; it removes the pin conflict.*

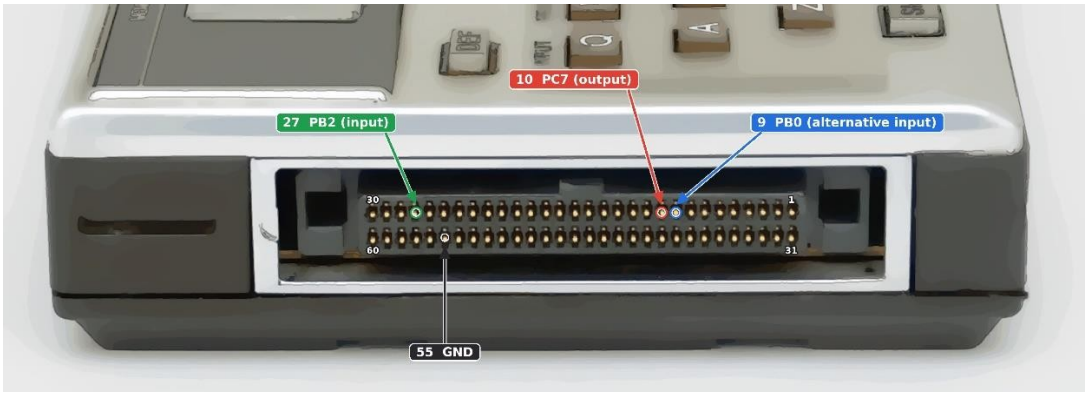
Features

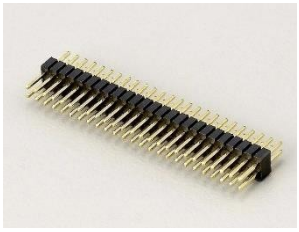
1. A BASIC installer is provided that installs the sending and receiving routines, SEROUT and SERIN, in the RAM of the PC-1500(A). The routines are written in machine language and occupy only about 200 to 250 bytes. In addition, two 255-byte buffers (TX and RX) are set up to hold the data to be transmitted or received. The routines work in "bit-banging" mode: the signals are generated and read bit by bit, at a low level, by the software itself rather than by a dedicated hardware circuit.

2. The following baud rates can be selected: **1200, 2400, 4800, 9600** or **19200 bps**. The default is 9600 bps, which lets the PC-1500(A) work with many popular UART devices; 4800 bps is a universal and reliable standard, while the slower 1200 and 2400 bps modes provide compatibility with vintage equipment. **19200 bps**, the highest rate the PC-1500(A) can handle (about 1.9 KB/s), uses a special fast version of the routines (v7.1), whose timing was calibrated on a real PC-1500 with an Arduino UNO (see the file CALIBRATION_19200.md). One installer, **v7.3**, offers all five rates; the installer v6.2 on the tape recording SERINOUT.WAV offers 1200 to 9600 bps.

3. This UART uses the common **8N1** format (8 data bits, no parity, 1 stop bit). RTS and CTS lines are not available. The signal levels are TTL/CMOS compatible. This is NOT an RS-232 with voltages of up to ±15 V: connecting the PC-1500(A) to an RS-232 connector will destroy its electronics. You can, however, use it with many +3.3 V/+5 V devices. UART operation has been confirmed with USB-UART cables (FTDI), various Sharp, Casio and NEC pocket computers, various sensors and modules, Arduino boards (5 V TTL levels) and also with an ESP32 (3.3 V CMOS levels) **without a level converter** in the ESP32 → PC-1500 direction: the PC-1500 accepts about 2.4 V or more as a high level, so the 3.3 V output of the ESP32 is enough. In the opposite direction (PC-1500 → ESP32) the voltage must be reduced so as not to damage the ESP32, whose inputs tolerate at most about 3.6 V. You can use, for example, a resistor divider (47 kΩ in series and 100 kΩ to GND give about 3.3 V; see diagram '*Connection 3*' below).

4. The PC-1500(A) has to be connected through its 60-pin side connector: **PC7** (pin 10) is always used for transmitting (PC-1500 output) and **PB2** (pin 27) for receiving (PC-1500 input). Connect these pins to the corresponding pins of the other device, i.e. the PC-1500(A) input to the output of the other device and the PC-1500(A) output to the input of the other device, through resistors of a few kΩ (I used 1 to 10 kΩ successfully) as shown on the *Connection 2* diagram. Since both devices have high-impedance inputs, only an extremely small current flows, so the resistor value is not critical and the resistor causes practically no voltage drop. To connect well to the Sharp's socket you can use a common 1.27 mm dual row pin header (2x30 pins).
   Use any of pins 52 to 55 to establish a ground connection between the two devices.





1.27 mm dual row pin header

NOTA BENE: *PB2 is used as the cassette input (CMT IN), so it cannot be used while the PC-1500(A) is docked in the printer/cassette interface CE-150. To make it possible to use this UART with the PC-1500(A) docked in the CE-150, the installer lets you choose the unused pin PB0 (pin 9) as the input instead of PB2. Using PB0 requires a simple modification inside the PC-1500(A), though (some units already have this connection). The serial lines also have to be brought out of the CE-150 interface. These changes are not difficult for anyone with basic electronics workshop skills, and the modification is described in the file Modification_for_use_with_CE-150_EN.pdf.*

5. After installation, the SEROUT and SERIN routines are called with CALL commands, with an optional parameter. Transmission or reception starts immediately: there is no need to open, select or close a port as is the case with the CE-158 interface or other computers with a hardware-implemented UART interface. The **send and receive commands are all you need** to handle UART communication.

6. This software UART is driven entirely by the PC-1500's CPU and not by a dedicated UART chip with its own buffers and handshake lines (such as the CE-158 interface). Communication is therefore packet-based, not continuous. This suits sensors, microcontrollers and request/response protocols that send data in bursts with pauses, but not continuous serial streams such as a terminal session. If you need a UART interface that runs continuously and at higher speeds, I recommend the CE-158X interface by Jeff Birt (www.soigeneris.com).

7. The installer does not fit in a **PC-1500 without a RAM extension (a CE-155 or another RAM module is needed)**, but it runs well on a bare **PC-1500A**. More information about RAM, the NEW commands and the addresses is given in the file HELP_NEW_memory.md.

How to use

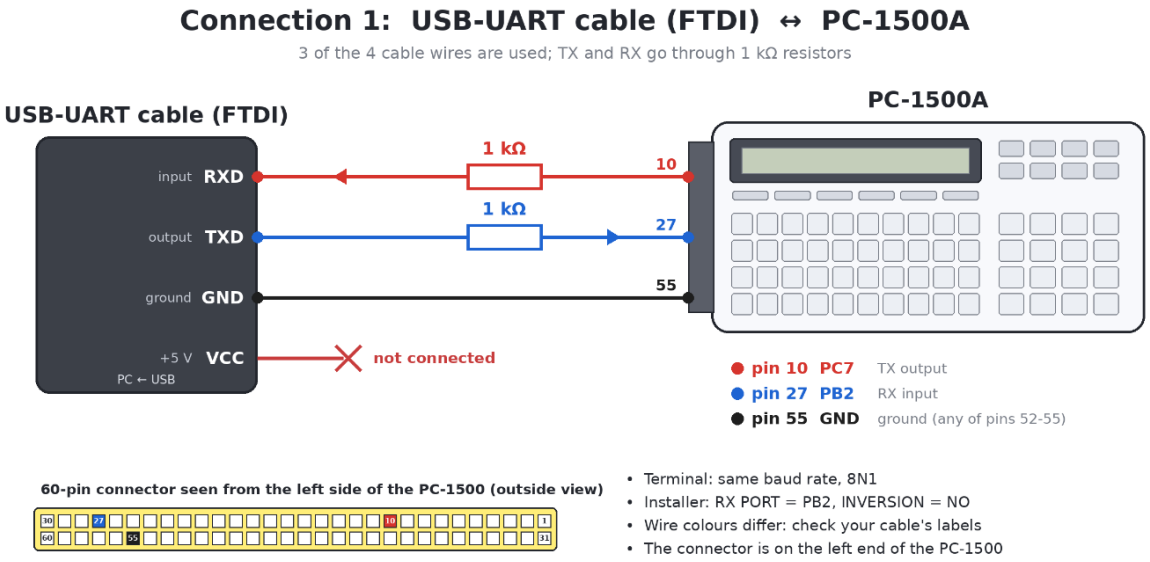
1. Switch off the PC-1500A and insert it into the CE-150 dock.
   Connect the sound source (the audio output of a computer or player) to the EAR jack of the CE-150 with an audio cable.

2. Switch on the PC-1500A and select PRO mode. Type:
   NEW0 [ENTER]
   NEW&4400 [ENTER] (or a different NEW&... value for a PC-1500(A) with a RAM module, see the file HELP_NEW_memory.md)
   CLOAD [ENTER]

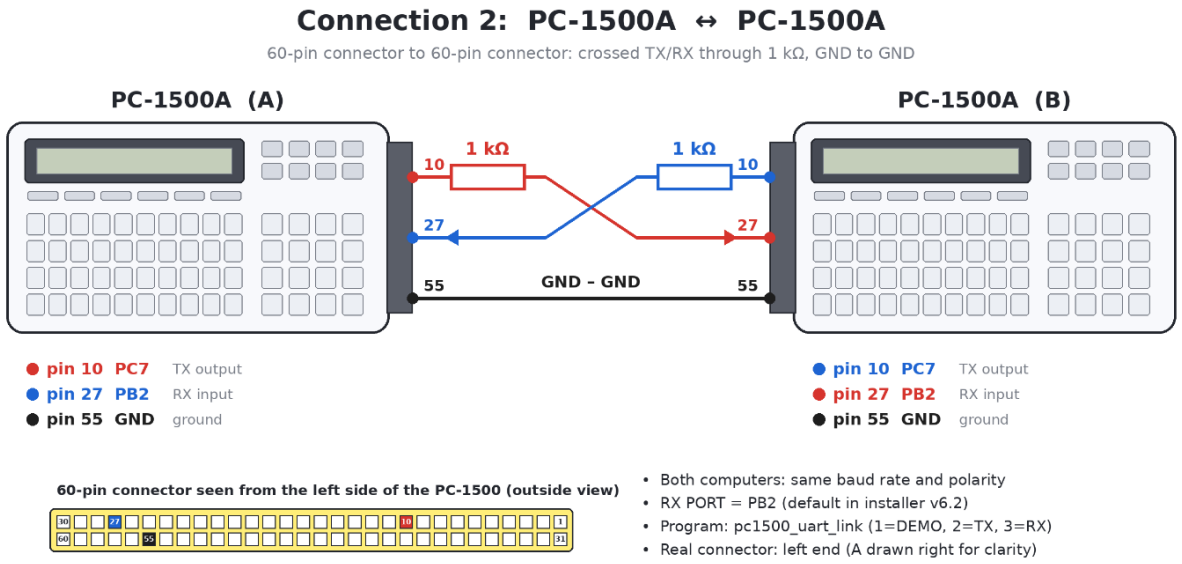
3. Start playing the file SERINOUT.WAV and wait until the installer has finished loading. This recording contains the installer v6.2 (1200 to 9600 bps). **For 19200 bps**, use the **installer v7.3** instead (file Basic/pc1500_uart_installer-v7.3.txt, all rates from 1200 to 19200 bps): after NEW&4400, type it in (or load it in another way) instead of CLOAD.

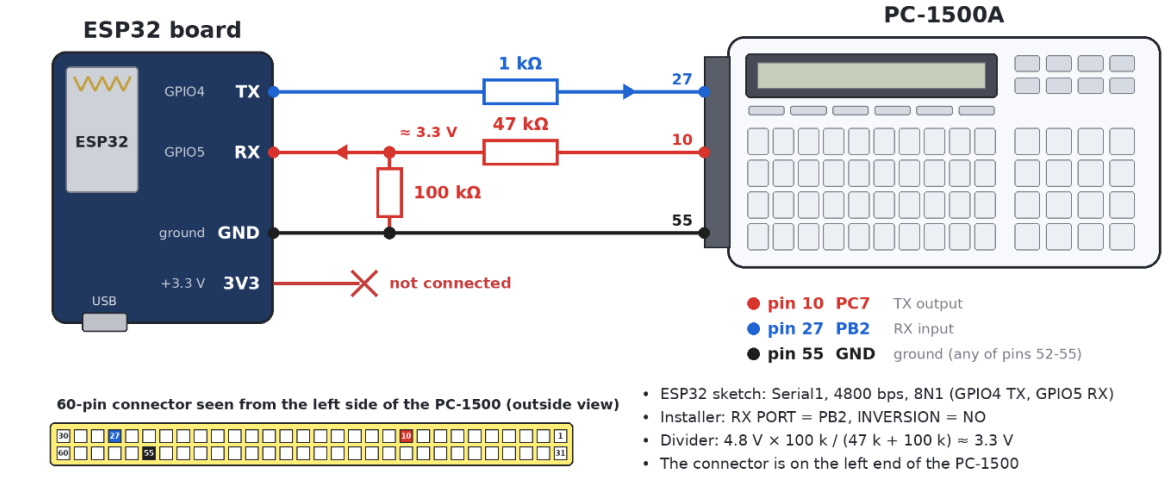
4. **If your PC-1500 and CE-150 are not modified**, switch off the PC-1500A and remove it from the CE-150.

5. Connect the PC-1500(A) to the other UART device. If you have no modification, connect the marked pins of the PC-1500A's 60-pin connector through 1–10 kΩ resistors (1 kΩ in the figures below) to the other UART device, which must use TTL or CMOS voltage levels. For a 3.3 V device such as the ESP32, use a resistor divider in the PC-1500 → device direction (Connection 3). Some possible connections:



- Terminal: same baud rate, 8N1
- Installer: RX PORT = PB2, INVERSION = NO
- Wire colours differ: check your cable's labels
- The connector is on the left end of the PC-1500





6. If the PC-1500 and CE-150 units have been modified, there is no need to use the wire connections as shown in *Connection 1..3* diagrams; instead, connect the UART device to the UART connector added to the CE-150 cassette/printer unit.

7. Switch on the PC-1500A, select RUN mode and type RUN [ENTER] to start the installer, which installs the SEROUT and SERIN routines. The installer v6.2 also contains useful sections that test TX and RX operation; for the installer v7.3 there is a separate tester (see point 9).

8. The installer asks a few questions; ENTER alone selects the default answer:

   - **Baud rate, installer v7.3**: it shows “1=1200 2=2400 3=4800 BPS” and asks “4/ENTER=9600 5=19200?”. Press **1** => 1200 bps, **2** => 2400 bps, **3** => 4800 bps, **4** or ENTER => 9600 bps, **5** => 19200 bps.
   - **Baud rate, installer v6.2** (SERINOUT.WAV): **1** => 1200 bps, **2** => 2400 bps, **4** or ENTER => 4800 bps, **9** => 9600 bps.
   - **Timing** (installer v7.3, 19200 bps only): “TIMING (ENTER=STD,1=SET)”. Press ENTER to use the standard, calibrated timing; 1 lets you enter the values found by a calibration of your own PC-1500 (see CALIBRATION_19200.md).
   - **RX port** (default PB2): without any modification of the PC-1500 and CE-150, the Software UART cannot be used while the PC-1500 is docked in the CE-150, and you must select the PB2 input (ENTER). If you have the modification, select the PB0 input by pressing 0.

- **Inversion** (default no): press 1 only if the other device uses inverted signal levels.

9. Once the setup process is complete, the installer v6.2 offers a TX test and an RX test. If the other UART device is not connected yet, press 0 to skip them; the RX test can be started later with RUN 530. The installer v7.3 has no test sections (together with the code of both versions of the routines, they would not fit in the memory of a PC-1500A): type NEW [ENTER] in PRO mode (the routines stay in memory), then load the **tester v7.3** (file Basic/pc1500_uart_test-v7.3.txt) and start it with RUN [ENTER]. It works at every baud rate.

10. The PC-1500 is now ready to work through the Software UART! You can use the SEROUT and SERIN routines by calling them with CALL commands. The commands for receiving and transmitting are simple and intuitive (note that they are different from the commands of the original Sharp CE-158):

    To send a string (packet) of data, use: CALL&nnnn,m (**CALL&40C5** on a PC-1500A without a RAM extension), where **nnnn** is the address of the SEROUT routine, shown by the installer; it depends on the RAM configuration of the computer*, and **m** is the length of the data packet to be sent (1–255). If the parameter **m** is not given, only the first byte of the TX buffer is sent. Before the call, put the data to be sent into the TX buffer with POKE commands; the buffer starts at an address that depends on the memory configuration* (&4300 on a PC-1500A without a RAM extension).

    To receive a string (packet) of data, use: CALL&nnnn,m (**CALL&4130** on a PC-1500A without a RAM extension), where **nnnn** is the address of the SERIN routine, shown by the installer; it depends on the RAM configuration, and **m** is the maximum length of the data packet to be received. If the parameter **m** is not given, a packet of any length is received into the RX buffer, up to 255 bytes. The actual length of the packet is stored in a dedicated byte (usually &41FF) and, if **m** was given, also returned as its new value. The SERIN routine listens for incoming bytes, decodes them and places them in the RX buffer. If no new character arrives within about 0.5 s, the routine stops listening, stores the packet length in the dedicated memory cell and returns to BASIC. The first character must arrive within about 30 s. The received data can then be read with PEEK commands from the RX buffer, which starts at an address that depends on the memory configuration* (&4200 on a PC-1500A without a RAM extension). The commands, addresses and buffers are the same at every baud rate, 19200 bps included.

11. The default baud rate is 9600 bps (4800 bps with the installer v6.2). To change the baud rate, run the installer again. If the other device uses inverted signal levels, one can select the INVERSION option during the installation.

12. To connect two PC-1500(A)s (*Connection 2*), run the link program (1=DEMO, 2=TX, 3=RX) on both of them: **link v2.0** (file Basic/pc1500_uart_link-v2.0.txt) after the installer v7.3, at any rate from 1200 to 19200 bps, or link v1.0 (link.wav) after the installer v6.2. Both computers must use the same baud rate and polarity.

13. If you no longer need the BASIC installer/tester, type **NEW** [ENTER] in PRO mode. This deletes the installer, but the routines stay in memory and the whole remaining memory is free for your BASIC program.

    **Check the addresses in the file HELP_NEW_memory.md, because they depend on the memory configuration of your PC-1500(A).*

Comments, bug reports, ideas and inquiries: alex.f4vts@gmail.com *(I may not be able to respond immediately).*

I have created the SERIN/SEROUT UART software by myself, with help from Claude Code.
It is based on the idea of the SAVER/LOADER program published in the Japanese magazine I/O (1983).